CORNELL ELECTRIC VEHICLES

CFD Analysis of UC26

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Quick Overview of CFD

- Navier-Stokes Equations
- Finite Volume Analysis

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

Basic steps

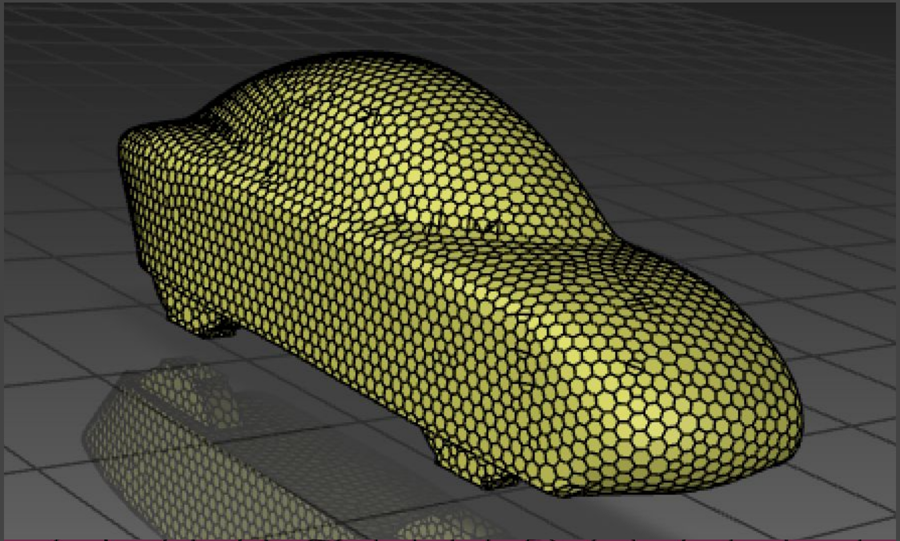
- Get geometry
- Create enclosure for geometry
- Create initial surface mesh
- Generate volume mesh
- Set up mathematical model
- Solve (hopefully) → post-processing

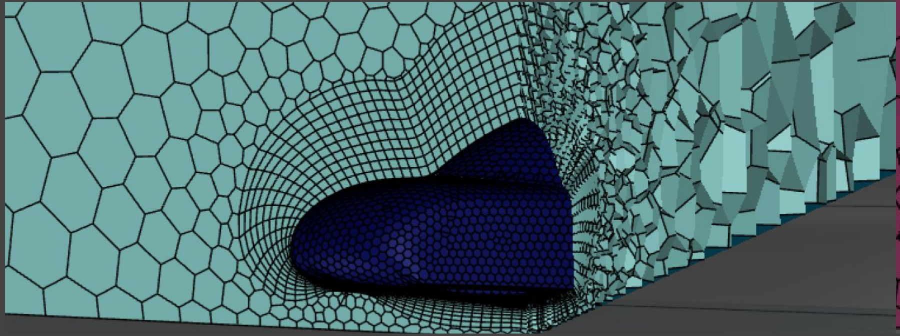
My mesh

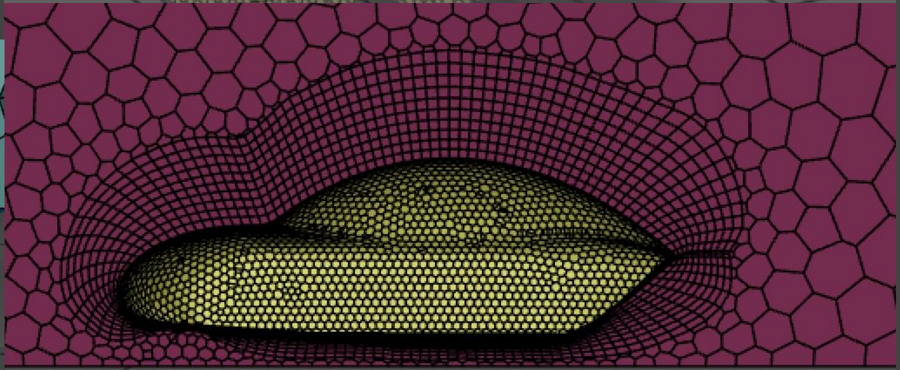
Orthogonal quality: 0.8758014

Skewness: 0.035590306

47531 cells





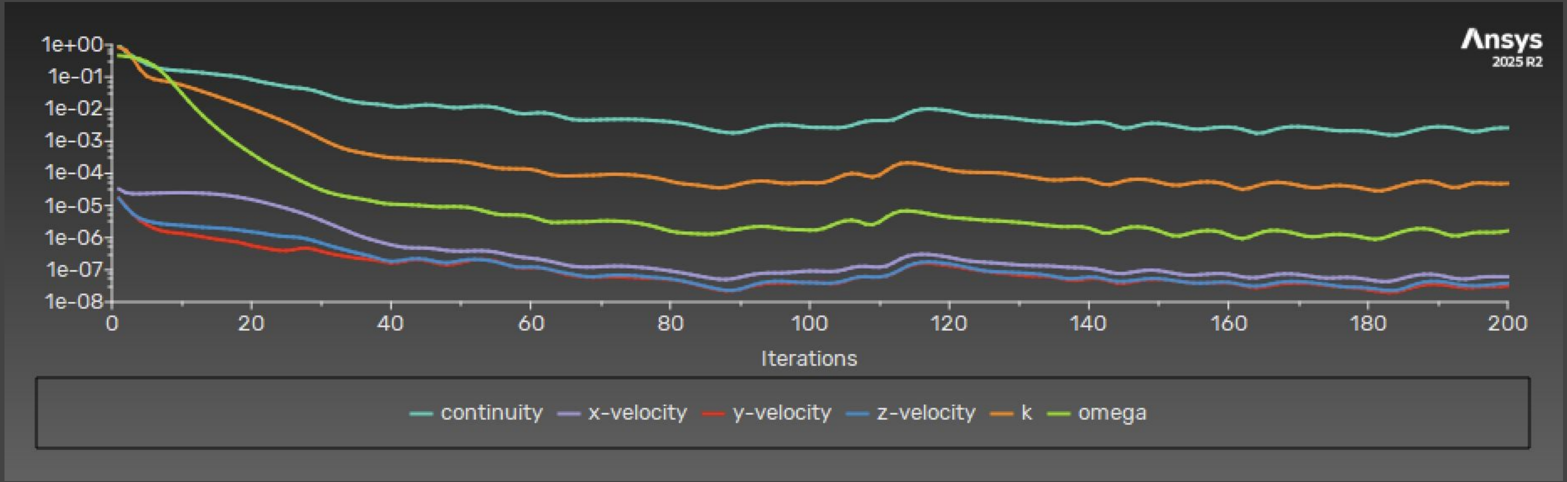


Solution

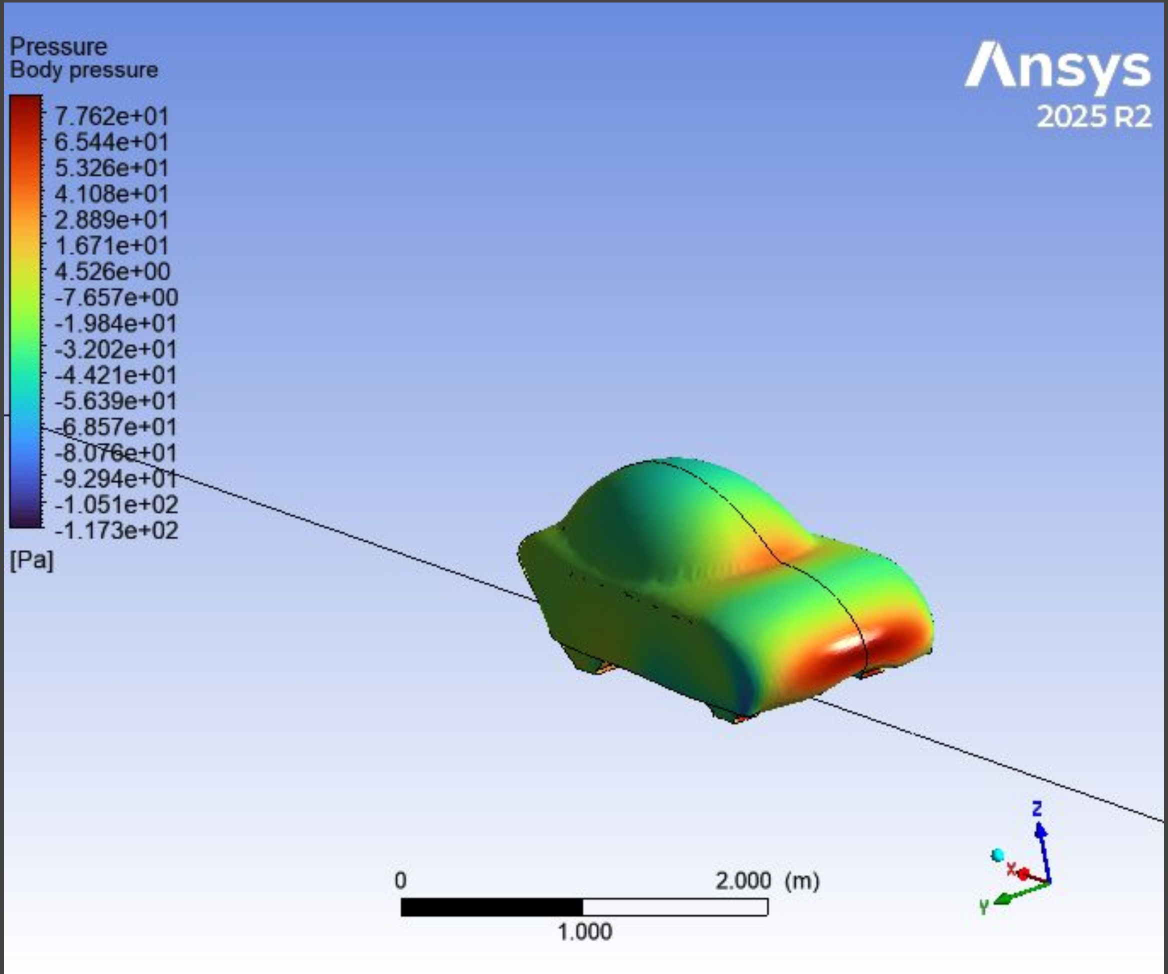
Had some convergence issues but results look pretty good

“These results look trustworthy” - lord Bhaskaran

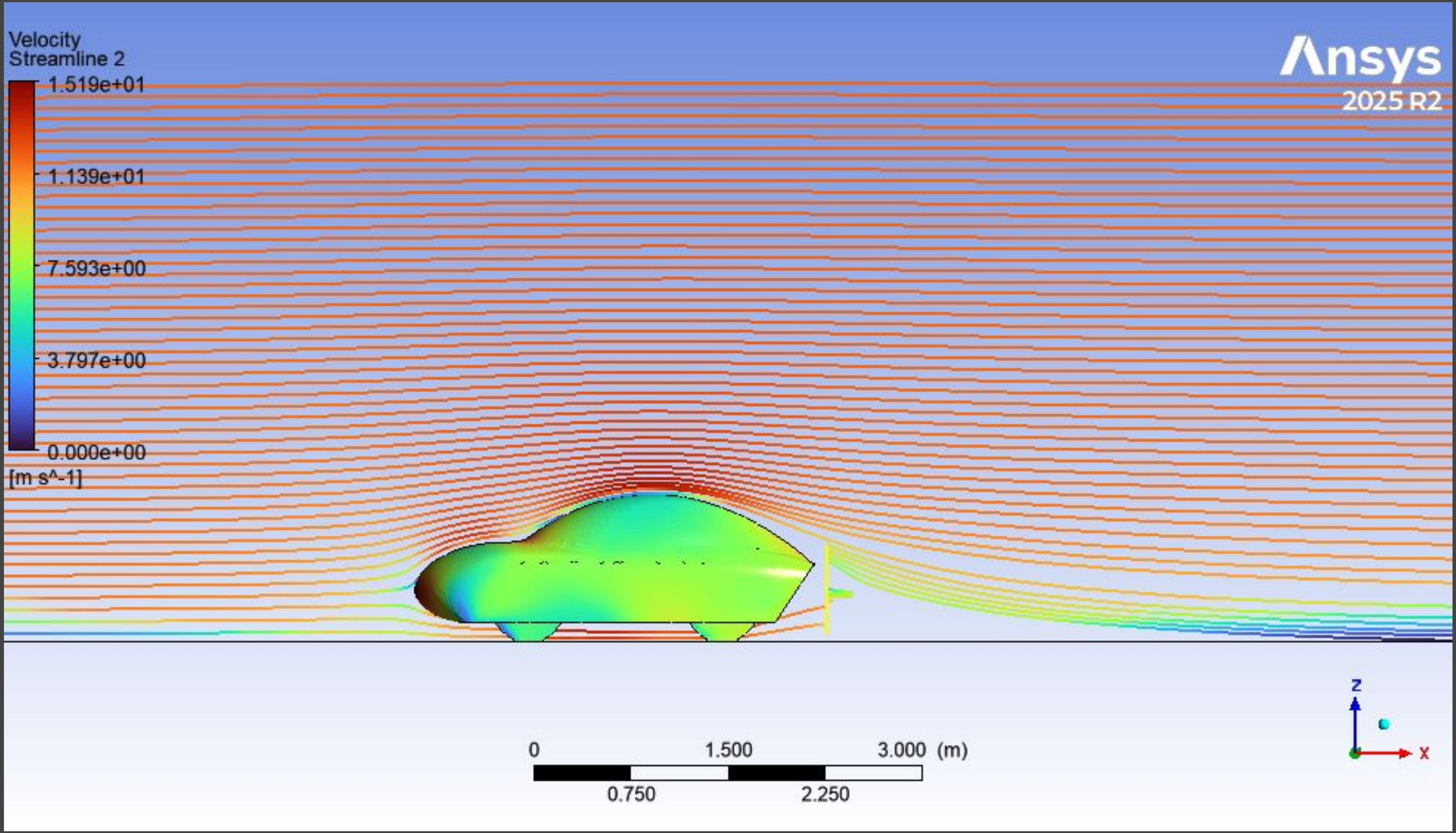
Cd = 0.15307361 (seems pretty realistic)

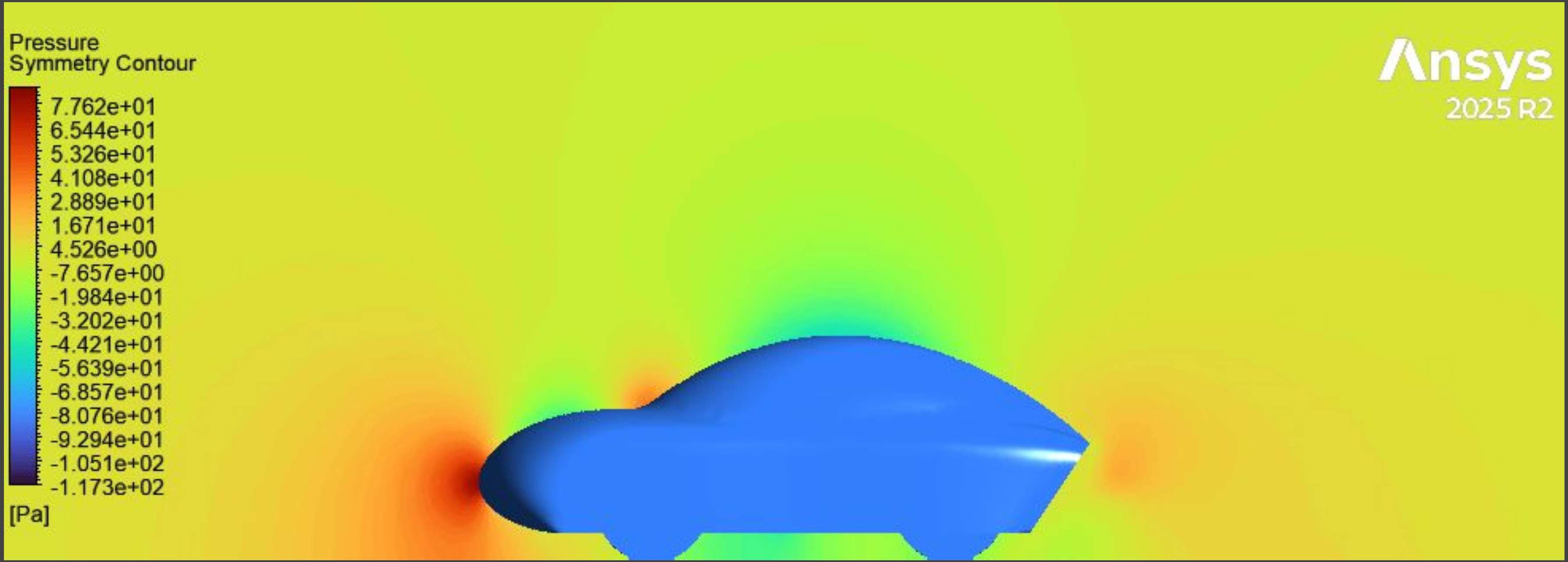


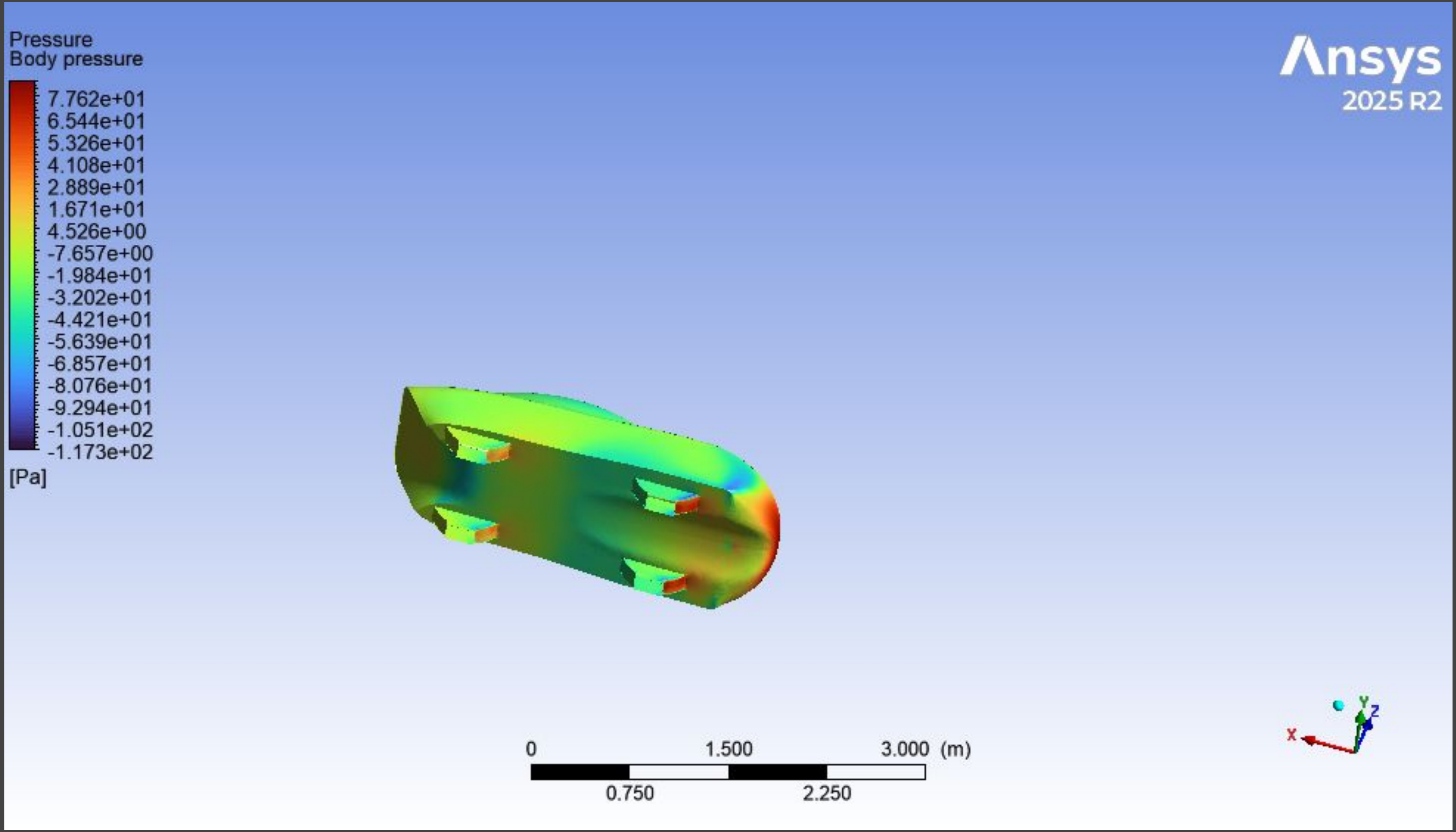
Images!



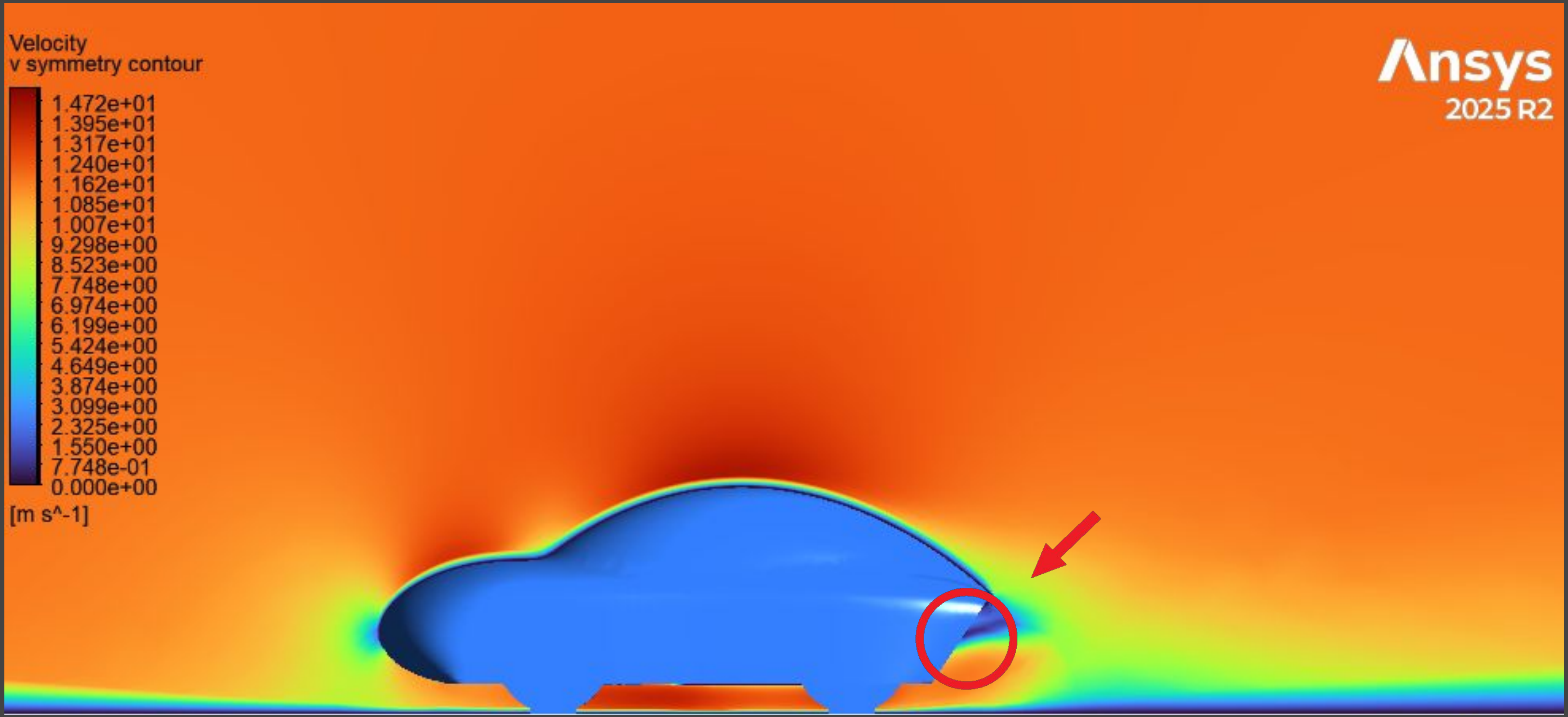
Pressure
Body pressure
7.762e+01
6.544e+01
5.326e+01
4.108e+01
2.889e+01
1.671e+01
4.526e+00
-7.657e+00
-1.984e+01
-3.202e+01
-4.421e+01
-5.639e+01
-6.857e+01
-8.076e+01
-9.294e+01
-1.051e+02
-1.173e+02
[Pa]
Ansys
2025 R2
0
2.000 (m)
1.000
Z
X
Y


Velocity
Streamline 2
1.519e+01
1.139e+01
7.593e+00
3.797e+00
0.000e+00
[m s^-1]
Ansys
2025 R2
0
1.500
3.000 (m)
0.750
2.250
Z
X


Pressure
Symmetry Contour
7.762e+01
6.544e+01
5.326e+01
4.108e+01
2.889e+01
1.671e+01
4.526e+00
-7.657e+00
-1.984e+01
-3.202e+01
-4.421e+01
-5.639e+01
-6.857e+01
-8.076e+01
-9.294e+01
-1.051e+02
-1.173e+02
[Pa]
Ansys
2025 R2

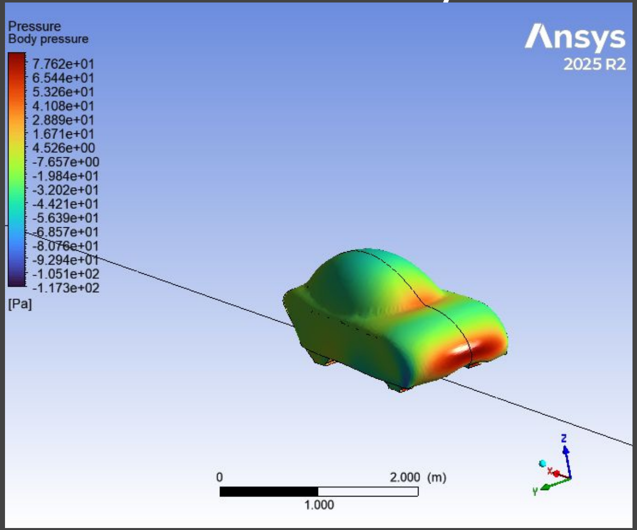


Pressure
Body pressure
7.762e+01
6.544e+01
5.326e+01
4.108e+01
2.889e+01
1.671e+01
4.526e+00
-7.657e+00
-1.984e+01
-3.202e+01
-4.421e+01
-5.639e+01
-6.857e+01
-8.076e+01
-9.294e+01
-1.051e+02
-1.173e+02
[Pa]
Ansys
2025 R2
0
1.500
3.000 (m)
0.750
2.250
X
Y
Z

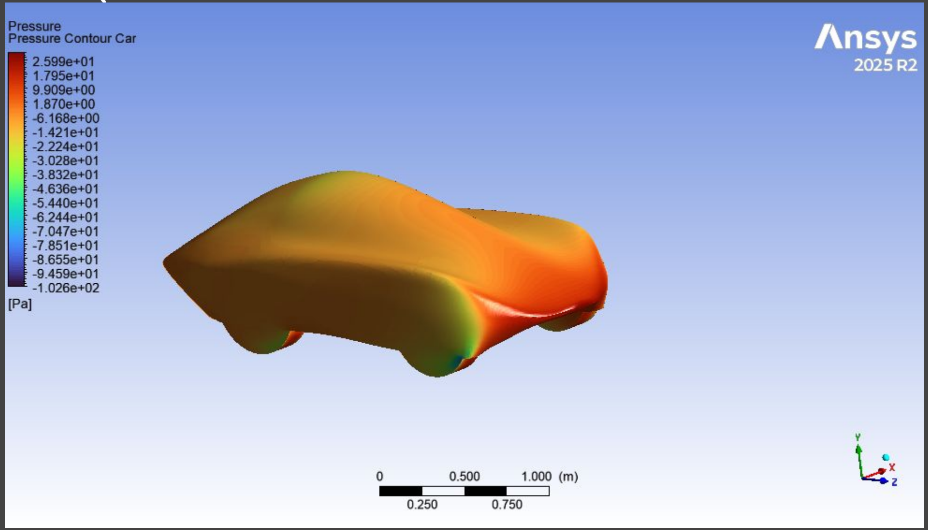

Velocity
v symmetry contour
1.472e+01
1.395e+01
1.317e+01
1.240e+01
1.162e+01
1.085e+01
1.007e+01
9.298e+00
8.523e+00
7.748e+00
6.974e+00
6.199e+00
5.424e+00
4.649e+00
3.874e+00
3.099e+00
2.325e+00
1.550e+00
7.748e-01
0.000e+00
[m s^-1]
Ansys
2025 R2

Comparison to previous analysis

- Able to get a more accurate drag coefficient
  - 0.15307361 vs 6.0420478
- Same continuity convergence issue :(



New



Old

What now?

- Attempt to fix continuity convergence issue
- **Mesh refinement**
- Analysis on different base plate designs with current aero
- Assist in aero design for UC27

Q&A